LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety and emergency devices for retraction and extension, typical u/c circuits, brief servicing aspects.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety and emergency devices for retraction and extension, typical u/c circuits, brief servicing aspects.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of typical u/c circuit diagrams from IB; identification of safety/emergency devices on schematic; discussion of servicing intervals and limitations.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear system?
(b) Name two safety devices used in landing gear retraction.
(c) Why is it critical to understand u/c circuits before servicing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Safety devices for retraction and extension • Emergency extension systems overview • Typical u/c circuit components • Hydraulic and mechanical actuation paths • Electrical control logic for gear	L & Ds	PPT, IB	15 min
(b)	• Servicing aspects and maintenance intervals • Inspection points for u/c systems • Common faults and troubleshooting • Lubrication requirements for u/c • Safety precautions during servicing	L & Ds	PPT, PDF	20 min
(c)	• Review of emergency extension procedures • Summary of circuit safety features • Key servicing limitations and warnings • Connection between safety devices and circuits • Preparation for next lesson on wheels	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the emergency extension system in a hydraulic failure scenario.
(b) Identify three critical inspection points for the u/c circuit during routine servicing.

12. Home Assignment and Follow Up	Prepare a brief note on the types of wheels and their description to be covered in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wheels types of wheels and description of typical main wheel, general repair to wheels.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min